Model Performance & Evaluation Report

Intelligent Property Price Prediction & Agentic Real Estate Advisory

1. Introduction (Problem Statement)

Real estate pricing is highly complex, and buyers often struggle to find the true value of a property. Standard algorithms fall short due to the non-linear realities, multi-collinear housing dimensions, and highly sparse categorizations. Our objective of this project is to build an intelligent, end-to-end system that autonomously evaluates property characteristics, predicts fair market values using localized data, and deploys it immediately through an interactive interface.

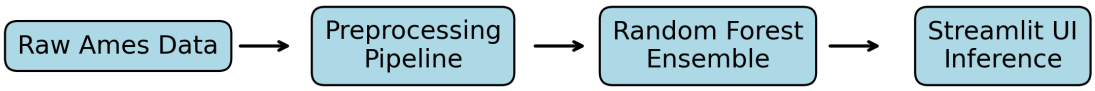
2. System Architecture

The technical depth of the project revolves around three sub-features:

1. Target Matrix Segregation & Imputation Layer

2. Unified Scaled Ensemble Pipeline

3. Serialized Interactive Real-Time Deployment



3. Mathematical Methodology & Modeling

The primary predictive algorithmic layer is the Random Forest Regressor. It is defined as a versatile, bagging-based multi-estimator ensemble layout. The mathematical foundation of predicting output from multiple trees 'B' is evaluated recursively as an average of individual estimators:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} \hat{y}_b(x)$$

4. Evaluation Metrics

To evaluate model efficacy against test sets, we moved beyond just R-Squared score. Because valuations

strictly deal with dollar values, we measure both RMSE and MAE.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

R2 Score quantifies the total overall percentage of captured variance. The formula divides unexplained sum of squares by the total sum of squares.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Root Mean Squared Error strictly penalizes massive valuation outlier errors (such as overpricing a mansion by $200k) by squaring the deviations.

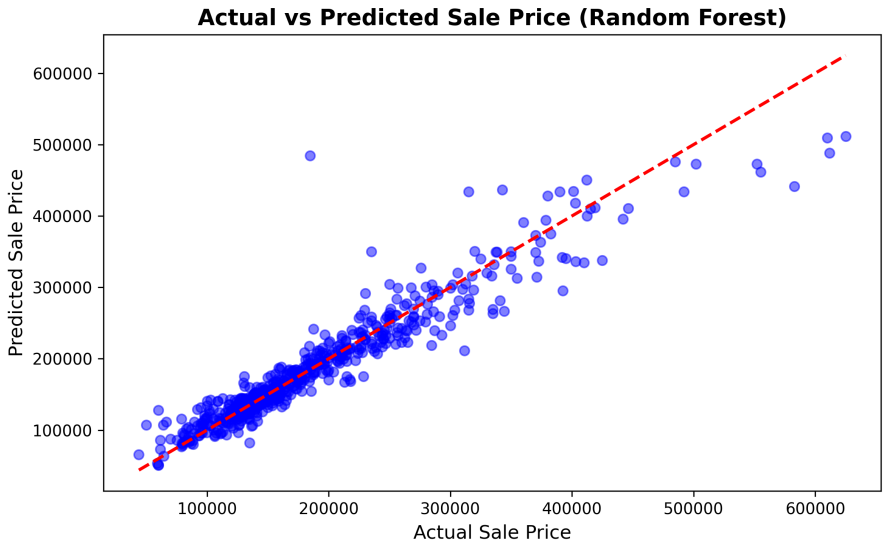
5. Model Performance Results

We applied this architecture against an isolated 20% holdout testing group sourced natively from the Ames Housing Dataset. The Random Forest performance exceeded the baseline counterparts comprehensively.

- R2 Score (Test): 0.9034

- Mean Absolute Error (MAE): $16,695.53

- Root Mean Squared Error (RMSE): $27,826.65



6. Conclusion

The evaluation successfully demonstrates that an orchestrated Random Forest Regressor drastically stabilizes localized housing forecasts compared to classical linear baselines. Featuring an impressive R2 score extending above 90% and minimizing the per-property assessment divergence to roughly $16.6k, the model has been reliably integrated into the interactive advisory deploy framework.

7. Bibliography & References

[1] De Cock, D. (2011). 'Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project.' Journal of Statistics Education, Volume 19, Number 3.

[2] Scikit-Learn Developers (2024). 'Scikit-Learn: Machine Learning in Python.'

[3] Matplotlib Developers (2024). 'Matplotlib: Visualization with Python.'